

Recovering a Sparse Matrix from Paired Observations

A Preliminary Problem Statement

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Abstract

summary goes here

1 Purpose

This document states one small problem completely, and solves it as far as algebra will carry it. The problem is the first step toward a larger programme: inference of vector time-series models in which the structure of the model is determined by the data rather than assumed beforehand. Nothing about that larger programme appears here. Later documents will add one element at a time.

The problem treated here is the following. We observe pairs of vectors, an input and an output, and we believe the output is a fixed linear function of the input, plus measurement error. We want the matrix of that linear function. We expect most of its entries to be zero, and we do not know beforehand which ones.

2 Bayes's theorem and the estimate we shall use

Let A denote a proposition and B a body of evidence. Bayes's theorem starts with the basic definition of conditional probability

$$P(A | B)P(B) = P(A, B) = P(B | A)P(A) \quad (2.1)$$

and derives the formula for updating beliefs given new evidence:

$$P(A | B) = \frac{P(B | A)P(A)}{P(B)} \quad (2.2)$$

$$P(B) = \int P(B | a)P(a)da \quad (2.3)$$

where the integral runs over every proposition a that might have produced the evidence, and becomes a sum when that set is countable.

The denominator $P(B)$ does not involve A . It is the same number for every candidate A under consideration. Consequently, if our purpose is to find the candidate A that makes $P(A | B)$ as large as possible, we may work with the numerator alone and never evaluate $P(B)$ at all:

$$\arg \max_A P(A | B) = \arg \max_A P(B | A)P(A) \quad (2.4)$$

This is maximum posterior likelihood estimation, and it is what this document does. The saving is not incidental. The integral defining $P(B)$ runs over every matrix that might have produced the data, and we have no way to evaluate it.

Two warnings apply to (2.4) and are taken up in Sections 4.2 and 5. First, the largest value of a probability density is not always attained; a density can increase without bound. Second, the position of the largest value depends on how the parameters are written, since a density is defined relative to a measure. Both matter here, and both are resolved by stating what we know rather than by mathematical device.

3 Notation

N	the dimension of the vectors; A is $N \times N$
T	the number of observation pairs
n	NT , the number of scalar measurements
q	N^2 , the number of cells in A
X_{jt}	component j of the input vector at observation t
Y_{it}	component i of the output vector at observation t
Z_{it}	the value of Y_{it} the matrix predicts
$\mathcal{S}$	the set of cells (i, j) at which $A_{ij} \neq 0$
M	the number of cells in $\mathcal{S}$
p	the probability that any one cell is non-zero
σ	the standard deviation of the measurement error
$R(A)$	the residual sum of squares

4 The space of matrices

The unknown A is a square array of N rows and N columns. Each cell holds a real number, positive or negative, or else holds nothing at all, which we write as zero. A candidate is therefore two things taken together: a choice of which cells are occupied, and a choice of value for each occupied cell.

The first choice is discrete. There are 2^q possible patterns, where $q = N^2$. The second is continuous, and once the pattern is fixed it involves M real numbers. Any prior over A must give both.

4.1 The pattern of occupied cells

We suppose that each of the q cells is occupied or not independently of the others, with the same probability p of being occupied. A pattern with M occupied cells then has probability

$$P(\mathcal{S} \mid p) = p^M (1 - p)^{q-M} \quad (4.1)$$

The second factor is necessary. Without it the expression does not sum to unity over the 2^q patterns, and the comparison between a sparse matrix and a dense one is not a comparison between probabilities.

We do not know p . The hyper-prior over it is taken to be a Beta distribution with parameters α and β ,

$$P(p) = \frac{p^{\alpha-1}(1-p)^{\beta-1}}{B(\alpha, \beta)} \quad (4.2)$$

where $0 \leq p \leq 1$. The mean of this distribution is $\alpha/(\alpha + \beta)$. Setting $\alpha = \beta = 1$ gives the uniform distribution on $[0, 1]$, which is the statement that nothing is known about p beyond its range. Choosing $\alpha/(\alpha + \beta)$ below one half states an expectation that the matrix is sparse without excluding a dense one, and the sum $\alpha + \beta$ governs how firmly that expectation is held. The general form is carried through the algebra below; Section 6.2 examines what influence it can have.

All algebra and integrals in this paper were derived and verified using the open-source symbolic mathematics package Maxima version 5.49.0.

4.2 The magnitudes of the occupied cells

For the value in an occupied cell we should like a statement that does not depend on the units in which X and Y are measured. The usual such statement is that the magnitude is equally likely to fall in any interval of fixed width on a logarithmic scale: as likely between 0.01 and 0.1 as between 10 and 100. Written as a density in a , that statement is proportional to $1/|a|$.

Taken over the whole real line, $1/|a|$ is not a probability density. Its integral diverges at zero and again at infinity,

$$\int_{\varepsilon}^1 \frac{da}{a} = -\log \varepsilon \xrightarrow{\varepsilon \rightarrow 0} \infty \quad (4.3)$$

$$\int_1^K \frac{da}{a} = \log K \xrightarrow{K \rightarrow \infty} \infty \quad (4.4)$$

so it has no normalising constant. Densities of this kind are used, and are sometimes harmless, but here they fail in two separate ways.

The posterior would have no maximum. Hold the pattern fixed. As will be shown in Section 7, the quantity to be minimised contains the term $\sum \log |A_{ij}|$, summed over occupied cells. Let one coefficient approach zero while the others are held fixed. The residual sum of squares is a continuous quadratic function of the coefficients, so it approaches the finite value it takes when that coefficient equals zero. The term $\log |A_{ij}|$, by contrast, decreases without bound. The quantity to be minimised therefore has no lower bound, and no maximising matrix exists.

The infimum is approached as the coefficient tends to zero, and at zero the candidate is no longer a member of the set being searched: a coefficient equal to zero means the cell is unoccupied, which is a different pattern with M smaller by one. More data does not repair this. Increasing n enlarges the term measuring fit relative to the others, but $\log |A_{ij}|$ still diverges.

Patterns of different size could not be compared. Suppose we avoided the first difficulty by integrating over the coefficient values instead of maximising over them. A density without a normalising constant is determined only up to an arbitrary positive factor c : the density $c/|a|$ expresses the same ignorance for every c . A pattern with M occupied cells then contributes c^M , and the comparison of a pattern of size M with one of size $M + 1$ is multiplied by c . Since c has no determined value, any conclusion whatever about the sparsity of A can be produced by choosing c suitably.

The general rule is this. A density without a normalising constant may be used when the number of factors it contributes is the same for every candidate, since the unknown constant then cancels from every comparison. For the coefficients of A that number is M , and M varies from candidate to candidate.

The remedy. We keep the logarithmic statement but confine it to a stated range of magnitudes. The analyst using this method knows, for the problem at hand, the smallest and the largest coefficient magnitudes worth entertaining. Call them $a_{\min}$ and $a_{\max}$, and write

$$L_A = \log \frac{a_{\max}}{a_{\min}} \quad (4.5)$$

The prior for one occupied cell is then

$$P(A_{ij}) = \frac{1}{2L_A |A_{ij}|} \quad (4.6)$$

where $a_{\min} \leq |A_{ij}| \leq a_{\max}$, the factor two accounting for the two signs. This integrates to unity. Both difficulties disappear: $\log |A_{ij}|$ is bounded below by $\log a_{\min}$, and the normalising constant is determined, so patterns of different size may be compared.

The invariance that recommended $1/|a|$ in the first place survives. A change in the units of the coefficients multiplies $a_{\min}$ and $a_{\max}$ by the same factor and leaves L_A unchanged. The quantity L_A has

a plain reading: it is the width of the admissible range of magnitudes, measured in natural logarithms. A range of ten powers of ten gives $L_A \approx 23$.

Collecting (4.1) and (4.6), the prior over the whole matrix, given p , is

$$P(A \mid p) = p^M (1-p)^{q-M} (2L_A)^{-M} \prod_{(i,j) \in \mathcal{S}} \frac{1}{|A_{ij}|} \quad (4.7)$$

5 The data and the measurement error

The data B consist of T pairs of N -dimensional real vectors. The inputs X are taken to be exact. All error is in the outputs. The matrix predicts

$$Z_{it} = \sum_{j=1}^N A_{ij} X_{jt} \quad (5.1)$$

and we suppose the discrepancies $Y_{it} - Z_{it}$ to be independent of one another, each drawn from a normal distribution of mean zero and standard deviation σ . Then

$$P(B \mid A, \sigma) = \prod_{i=1}^N \prod_{t=1}^T \frac{1}{\sigma \sqrt{2\pi}} \exp\left(-\frac{(Z_{it} - Y_{it})^2}{2\sigma^2}\right) = (2\pi)^{-n/2} \sigma^{-n} \exp\left(-\frac{R(A)}{2\sigma^2}\right) \quad (5.2)$$

where

$$R(A) = \sum_{i=1}^N \sum_{t=1}^T (Y_{it} - Z_{it})^2 \quad (5.3)$$

and $n = NT$. The factors of $\sqrt{2\pi}$ contribute $(2\pi)^{-n/2}$, which does not involve A , p or σ . It is the same number for every candidate and may be dropped throughout without affecting which candidate is best.

5.1 The scale of the error

We do not know σ . The natural statement of ignorance about a scale is again the logarithmic one, proportional to $1/\sigma$, and again it has no total mass.

Here the consequence differs from the one in Section 4.2. Every candidate matrix carries exactly one factor of the prior for σ , so the arbitrary constant does cancel. The difficulty lies elsewhere, and it is clearest in the integrand itself. Combining the prior $1/\sigma$ with the likelihood (5.2), and setting aside factors that do not involve σ , the integral to be performed is

$$\int_0^\infty \sigma^{-(n+1)} \exp\left(-\frac{R}{2\sigma^2}\right) d\sigma \quad (5.4)$$

The factor $\sigma^{-(n+1)}$ is by itself not integrable at the origin. What makes (5.4) converge there is the exponential, which drives the integrand to zero as σ decreases. The reach of that suppression is governed by R . The integrand rises to a single maximum at

$$\sigma^2 = \frac{R}{n+1} \quad (5.5)$$

and falls away on both sides. As R decreases, the maximum moves toward the origin and its height, proportional to $R^{-(n+1)/2}$, grows without limit. The suppression that made the integral finite is confined to an ever smaller neighbourhood of zero, exposing more and more of the singularity in $\sigma^{-(n+1)}$. Performing the integral,

$$\int_0^\infty \sigma^{-(n+1)} \exp\left(-\frac{R}{2\sigma^2}\right) d\sigma = \frac{1}{2} \Gamma\left(\frac{n}{2}\right) \left(\frac{2}{R}\right)^{n/2} \quad (5.6)$$

which increases without bound as R approaches zero. The divergence occurs at $\sigma = 0$, and it appears in the answer as the factor $R^{-n/2}$. Stated in words: as the fit improves, an ever wider range of very small error scales becomes consistent with the data, and the accumulated weight of those scales has no upper limit.

The divergence is not a defect of the algebra. It is the formulation reporting that nothing has been said about how small the error can be. With $1/\sigma$ taken over $(0, \infty)$, a residual of 10^{-12} is admitted as evidence of an instrument accurate to 10^{-12} , and no part of the model contradicts that reading. Stating s_0 contradicts it.

Whenever the data can be fitted exactly, therefore, the exact fit dominates every other candidate however large its pattern.

Whether an exact fit is available depends on N and T . Row i of A is constrained only by row i of Y , which supplies T equations, and it contains as many free coefficients as that row has occupied cells. Two regimes are worth recording:

	$N = 10, T = 20$	$N = 200, T = 10$
cells in A	100	40,000
measurements n	200	2,000
equations per row	20	10
coefficients available per row	10	200
exact fit possible	no	yes, easily

In the first regime every row is over-determined and no exact fit exists. In the second, a pattern with ten occupied cells in each row reproduces the data exactly, and there are a great many such patterns. The second regime is the one we must be able to handle, so the difficulty cannot be set aside.

The remedy is the one already applied to the coefficients. The analyst who can state the range of credible coefficient magnitudes can equally state the range of credible noise levels: s_0 , below which the instrument cannot measure, and s_1 , above which the data would be worthless. Writing

$$L_\sigma = \log \frac{s_1}{s_0} \quad (5.7)$$

$$P(\sigma) = \frac{1}{L_\sigma \sigma} \quad (5.8)$$

$$s_0 \leq \sigma \leq s_1 \quad (5.9)$$

gives a proper density, and, as Section 6.3 shows, a finite result at $R = 0$. The statement $\sigma \geq s_0$ is a statement about the apparatus, not a mathematical convenience: it says that the measurements are not exact, and how inexact they are known to be.

6 The posterior for A , p and σ

Multiplying the prior (4.7), the hyper-priors on p and σ , and the likelihood (5.2), and discarding the constant $(2\pi)^{-n/2}$, the posterior for the three unknowns together is proportional to

$$P(A, p, \sigma \mid B) \propto p^M (1-p)^{q-M} \cdot \frac{(2L_A)^{-M}}{\prod_S |A_{ij}|} \cdot \frac{1}{L_\sigma \sigma} \cdot \sigma^{-n} \exp\left(-\frac{R(A)}{2\sigma^2}\right) \quad (6.1)$$

The quantities p and σ are not of interest in themselves. We remove them by integration, which is the step that makes the estimate of A depend on the data alone and not on values we should otherwise have to guess.

6.1 Removing p

Only the first factor of (6.1) involves p . With the hyper-prior (4.2),

$$\frac{1}{B(\alpha, \beta)} \int_0^1 p^{M+\alpha-1} (1-p)^{q-M+\beta-1} dp = \frac{B(M+\alpha, q-M+\beta)}{B(\alpha, \beta)} \quad (6.2)$$

At $\alpha = \beta = 1$ this reduces to

$$B(M+1, q-M+1) = \frac{M! (q-M)!}{(q+1)!} = \frac{1}{(q+1) \binom{q}{M}} \quad (6.3)$$

whose negative logarithm is $\log \binom{q}{M} + \log(q+1)$. The first term is the number of natural units of information required to say which M of the q cells are occupied. This is Occam's razor in a form that can be computed: a pattern must reduce the residual by enough to account for the information needed to describe it.

In either case the result depends on the pattern only through its size. Write

$$\Pi(M) = -\log \frac{B(M+\alpha, q-M+\beta)}{B(\alpha, \beta)} \quad (6.4)$$

for the contribution of the pattern to the quantity to be minimised.

6.2 How much the pattern term can do

Since Π depends only on M , its whole effect is contained in the amount by which it grows when one more cell is occupied. That increment has a simple closed form. The ratio of successive values of the Beta function is

$$\frac{B(M+\alpha, q-M+\beta)}{B(M+1+\alpha, q-M-1+\beta)} = \frac{q-M-1+\beta}{M+\alpha} \quad (6.5)$$

so that

$$\Pi(M+1) - \Pi(M) = \log \frac{q-M-1+\beta}{M+\alpha} \quad (6.6)$$

Two consequences follow, and both matter for what a Beta hyper-prior can accomplish.

First, the increment depends on β only through its logarithm. To raise the charge for one more occupied cell by D natural units above its value under the uniform hyper-prior requires β to grow by a factor of about e^D . A hyper-prior that is merely sceptical of dense matrices therefore changes the increment very little. With $q = 81$ and $M = 27$, the increment is 0.657 at $\alpha = \beta = 1$, and 1.281 at $\alpha = 1$, $\beta = 50$, a hyper-prior whose mean is $1/51$. Supplying a further six natural units would require β of order 10^4 , which asserts that the expected number of occupied cells in the whole matrix is less than one hundredth.

Second, the increment falls as M rises, so the pattern term resists the first few occupied cells more than the last. Its total contribution across the whole matrix is bounded by $q \log q$, of order four hundred natural units for $q = 81$, whereas the term measuring fit varies over hundreds of units for a single pattern. The hyper-prior on p is therefore a refinement, not a control. Anything that must be prevented has to be prevented elsewhere.

The posterior mean of p , should it be wanted, is $(M + \alpha)/(q + \alpha + \beta)$.

6.3 Removing σ

The remaining factors involving σ give

$$I(R) = \frac{1}{L_\sigma} \int_{s_0}^{s_1} \sigma^{-(n+1)} \exp\left(-\frac{R}{2\sigma^2}\right) d\sigma \quad (6.7)$$

The substitution $u = 1/\sigma^2$ turns this into an incomplete gamma integral. With $\Gamma(m, x) = \int_x^\infty u^{m-1} e^{-u} du$ and $m = n/2$,

$$I(R) = \frac{1}{2L_\sigma} \left(\frac{2}{R}\right)^{n/2} \left[\Gamma\left(\frac{n}{2}, \frac{R}{2s_1^2}\right) - \Gamma\left(\frac{n}{2}, \frac{R}{2s_0^2}\right) \right] \quad (6.8)$$

Two limits show what the bounds accomplish. If R is comfortably larger than $2s_0^2$, the second incomplete gamma function is negligible and the first is close to $\Gamma(n/2)$, so

$$I(R) \approx \frac{\Gamma(n/2)}{2L_\sigma} \left(\frac{2}{R}\right)^{n/2} \quad (6.9)$$

which is (5.6), the result obtained from the unbounded prior divided by L_σ . The bounds cost nothing when the fit is ordinary. If instead R approaches zero, the factor $R^{-n/2}$ is cancelled exactly by the behaviour of the bracket, and

$$\lim_{R \rightarrow 0} I(R) = \frac{1}{nL_\sigma} (s_0^{-n} - s_1^{-n}) \quad (6.10)$$

a finite number. A perfect fit is now merely very good, not infinitely good, and it must still account for the size of its pattern.

7 The criterion for the matrix

Collecting (6.1), (6.2) and (6.8), the posterior for the matrix alone is proportional to

$$P(A \mid B) \propto e^{-\Pi(M)} \cdot \frac{(2L_A)^{-M}}{\prod_{\mathcal{S}} |A_{ij}|} \cdot I(R(A)) \quad (7.1)$$

It is convenient to work with the negative logarithm, which is to be made as small as possible. Discarding terms that do not involve A ,

$$F(A) = \frac{n}{2} \log R(A) + \sum_{(i,j) \in \mathcal{S}} \log |A_{ij}| + M \log(2L_A) + \Pi(M) - \log \Delta(R) \quad (7.2)$$

where

$$\Delta(R) = \Gamma\left(\frac{n}{2}, \frac{R}{2s_1^2}\right) - \Gamma\left(\frac{n}{2}, \frac{R}{2s_0^2}\right) \quad (7.3)$$

The five terms have plain meanings. The first measures how badly the matrix fits the data. The second and third concern the occupied coefficients: how large they are, and how many. The fourth, given by (6.4), is the information required to specify which cells are occupied. The fifth is nearly constant over the ordinary range of R , by (6.9), and takes effect only when the fit becomes so good as to be implausible given the stated precision of the measurements.

Away from that extreme, $\Delta(R)$ may be replaced by $\Gamma(n/2)$ and dropped, so that the working criterion is

$$F(A) = \frac{n}{2} \log R(A) + \sum_{(i,j) \in \mathcal{S}} \log |A_{ij}| + M \log(2L_A) + \Pi(M) \quad (7.4)$$

One restriction on the use of (7.2) must be stated at once. It is a sound guide to the coefficient values within a pattern that has been fixed in advance, and Section 8 uses it for that. It is not a sound guide to the choice between patterns of different size, and Section 9 explains why and gives the quantity that is.

8 The coefficients for a given pattern

Suppose the pattern $\mathcal{S}$ is fixed. The terms in (7.4) that involve M are then constant, and we minimise over the values in the occupied cells. Write

$$e_{ij} = \sum_{t=1}^T (Y_{it} - Z_{it}) X_{jt} \quad (8.1)$$

the inner product of the residuals in row i with the j th input series, so that $\partial R / \partial A_{ij} = -2e_{ij}$. Setting the derivative of (7.4) to zero gives, for every occupied cell,

$$e_{ij} = \frac{R(A)}{n A_{ij}} \quad (8.2)$$

Ordinary least squares would give $e_{ij} = 0$. The estimate here departs from least squares by a quantity of order R/n , which is small when the measurements are many. The direction of the departure is toward smaller coefficients, since the density (4.6) places more of its value at small magnitudes.

That shrinkage deserves a word of caution. It is a consequence of maximising a density rather than integrating one, and its size depends on the fact that we chose to write the prior as a density in A_{ij} rather than in $\log |A_{ij}|$. It is not a statement about the data. Since it is of order R/n , it is negligible in the cases of interest, and (8.2) is retained as the estimate of the coefficients throughout.

9 Comparing patterns of different size

Equation (7.2) is not adequate for choosing between patterns, and the reason is arithmetical. Occupying one more cell increases Π by (6.6), which is under one natural unit in the middle of the range, and increases $M \log(2L_A)$ by $\log(2L_A)$, which is about two. Against that, one more coefficient reduces R , and the first term of (7.2) carries the factor $n/2$. When T is small the reduction is substantial and the two charges above do not approach it. Taken as a guide to the size of the pattern, (7.2) occupies as many cells as it is permitted to.

What is missing is any account of the freedom that the extra coefficient uses up. Maximising over the coefficients records the height of the posterior at its best point and disregards how wide the region of good fit is. A pattern with more coefficients reaches a higher point, but over a region so much narrower that its total weight may be smaller. The comparison must therefore be made between integrals, not between heights.

9.1 The integral over the coefficients

Fix the pattern $\mathcal{S}$ and let $\hat{A}$ be the least-squares point within it, $\hat{R}$ the residual sum of squares there. The residual is quadratic, so

$$R(A) = \hat{R} + d'Gd \quad (9.1)$$

where d collects the departures of the M coefficients from $\hat{A}$ and G is block diagonal with blocks $G_i = X_{\mathcal{S}_i}X'_{\mathcal{S}_i}$, one for each row. The factor $\prod 1/|A_{ij}|$ varies slowly across the region where the exponential is appreciable, so it may be evaluated at $\hat{A}$ and taken outside. The remaining integral is Gaussian, and gives

$$\int \exp\left(-\frac{R(A)}{2\sigma^2}\right) dA = \exp\left(-\frac{\hat{R}}{2\sigma^2}\right) (2\pi\sigma^2)^{M/2} \prod_{i=1}^N |G_i|^{-1/2} \quad (9.2)$$

This requires each G_i to be non-singular, which holds when no row has more than T occupied cells. A row with more than T occupied cells reproduces its own data exactly and leaves directions in which the residual does not change at all; such patterns are excluded here.

9.2 The integral over σ , once more

The factor σ^M from (9.2) multiplies the $\sigma^{-(n+1)}$ of Section 6.3, so the exponent becomes $M - n - 1$. Writing

$$\nu = \frac{n - M}{2} \quad (9.3)$$

and repeating the substitution $u = 1/\sigma^2$,

$$\int_{s_0}^{s_1} \sigma^{M-n-1} \exp\left(-\frac{\hat{R}}{2\sigma^2}\right) d\sigma = \frac{1}{2} \left(\frac{2}{\hat{R}}\right)^\nu \left[\Gamma\left(\nu, \frac{\hat{R}}{2s_1^2}\right) - \Gamma\left(\nu, \frac{\hat{R}}{2s_0^2}\right) \right] \quad (9.4)$$

Setting $M = 0$ recovers (6.8) exactly.

9.3 The criterion for the pattern

Collecting the factors and taking the negative logarithm, the pattern is chosen by minimising

$$\Phi(\mathcal{S}) = \nu \log \hat{R} + \sum_{(i,j) \in \mathcal{S}} \log |\hat{A}_{ij}| + \frac{1}{2} \sum_{i=1}^N \log |G_i| - \frac{M}{2} \log 2\pi + M \log(2L_A) + \Pi(M)$$

(9.5)

away from the extreme of a near-exact fit, where the bracket in (9.4) must be retained in place of $\Gamma(\nu)$.

Two differences from (7.4) carry the whole effect. The factor on $\log \hat{R}$ has fallen from $n/2$ to $(n - M)/2$, which is the count of measurements less the count of quantities fitted to them. And two new terms appear, $\frac{1}{2} \sum \log |G_i|$ and $-\frac{1}{2} M \log 2\pi$, which together measure how narrow the region of good fit has become. A coefficient that the data determine sharply makes $|G_i|$ large and raises Φ ; a coefficient that the data barely constrain adds little. That is the accounting for freedom which (7.2) omits.

9.4 A note on maximising against integrating

The step taken in this section is narrow. Within a fixed pattern the coefficients are still estimated by maximising, through (8.2); only the comparison between patterns of different size is made by integration. It is worth recording that the wider step may repay the effort.

The maximum of a density is a fragile quantity in two respects. It is not preserved under a change of variable: writing the same distribution as a density in $\log |a|$ rather than in a moves the maximum, while the integral over any region is unaffected. And for a density that decreases monotonically, the maximum sits at an endpoint whatever the parameters may be. The exponential density $\lambda e^{-\lambda x}$ on $x \geq 0$ attains its maximum at $x = 0$ for every λ , so the location of the maximum carries no information about λ at all. The prior (4.6) is of exactly this character, which is why Section 4.2 had to bound it and why (8.2) shrinks the coefficients slightly.

Wherever a conclusion rests on comparing regions of the parameter space rather than on locating one point in it, the integral is the quantity with a stable meaning. The present document integrates over p , over σ , and now over the coefficients when patterns are compared. As the model is enlarged in later documents, results that appear anomalous should be tested by integrating rather than maximising before any other explanation is sought.

10 Two numerical illustrations

The algebra above says how a matrix is to be scored. Whether the score selects the right matrix is a separate question, and this section answers it on manufactured data where the truth is known.

Two matrices are used. In the first, the occupied cells form three blocks of size three along the diagonal. In the second, the same coefficient values are moved to three columns of each row chosen at random, so

that no arrangement remains. Both matrices therefore have the same number of occupied cells, the same values, and the same count of three per row. They differ only in where those cells sit. The two cases of a given trial share the same inputs and the same measurement errors, so the comparison between them is paired.

Nothing in the prior refers to blocks. All $q = 81$ cells are treated alike, and the rows are fitted separately. If the block structure is recovered, it is recovered from the data.

10.1 Construction

dimension and observations	$N = 9, \quad T = 6, \quad q = 81, \quad n = 54$
occupied cells	three per row, so $M = 27$
coefficient magnitudes	drawn log-uniformly on $[0.40, 3.00]$, signs equally likely
stated coefficient range	$a_{\min} = 0.20, \quad a_{\max} = 5.0, \quad L_A = 3.219$
measurement error	$\sigma = 0.10$
stated error range	$s_0 = 0.02, \quad s_1 = 10, \quad L_\sigma = 6.215$
hyper-prior on p	$\alpha = \beta = 1$, the uniform case
inputs	X_{jt} drawn independently from the standard normal

The stated range $[a_{\min}, a_{\max}]$ lies strictly outside the range from which the true coefficients are drawn. That separation matters. A floor placed at the edge of the truth rejects those coefficients that the measurement error happens to push below it, and the method then misses cells it would otherwise find.

The proportions of N and T are the point of the exercise. Each row supplies six equations for nine candidate coefficients, so no row can be estimated without first deciding which cells it occupies. Occupying three of them leaves three degrees of freedom. Rows are restricted to at most $T - 1 = 5$ occupied cells, which excludes exact fits and keeps every G_i of Section 9 non-singular.

10.2 The search

Section 4 observed that a matrix of order N admits 2^{N^2} patterns of occupied cells. For $N = 9$ that is 2^{81} , or about 2.4×10^{24} , so the patterns cannot be examined one by one and some procedure is needed. This section states the one used.

Three properties of the problem make it tractable. The first is the restriction of Section 10.1 to at most $T - 1 = 5$ occupied cells in a row, which is required in any case for the residual to retain any degrees of freedom and for the matrices G_i of Section 9 to be non-singular. A row then admits

$$\sum_{k=0}^5 \binom{9}{k} = 382 \tag{10.1}$$

subsets of columns rather than $2^9 = 512$. The second is that the rows are fitted separately, so the fit of a row depends on that row alone. Enumerating every admissible subset of every row therefore takes $9 \times 382 = 3438$ least-squares fits, and this is done once. For each row and each size k the subset of that size with the lowest cost is retained, which leaves six numbers per row.

The third property requires more care, because Φ does not separate across rows. It couples them twice: through $\log \hat{R}$, which involves the sum of the residuals of all nine rows, and through $\Pi(M)$, which involves the total count of occupied cells. Holding σ fixed removes the first coupling, since the joint posterior of (A, σ) contains $\hat{R}/2\sigma^2$ rather than its logarithm, and a sum inside a logarithm becomes a sum of separate terms. What remains is a problem of allocating a total of M occupied cells among nine rows so as to minimise a sum of per-row costs, and that is solved exactly by dynamic programming across the rows, indexed by the number of cells allocated so far. One pass yields the best allocation for every total M at once.

The procedure is therefore: fix σ at each of 240 values spaced logarithmically across $[s_0, s_1]$; for each, run the dynamic programme and record the pattern it gives for every M ; discard duplicates; evaluate Φ exactly on every distinct pattern that survives; and take the lowest. In the trial reported below this examined 862 distinct patterns in the block case and 790 in the scattered case. The coefficient values of the winner are then moved from least squares to the stationary point of (8.2).

What this does and does not guarantee should be stated plainly. For any fixed σ the dynamic programme returns the exact minimiser of the separable form, so nothing is lost within a given σ . Across σ the grid is finite, and Φ itself is minimised only over the patterns the grid happens to produce. Some 1.7×10^{23} patterns satisfy the per-row restriction, of which a few hundred are examined, so the procedure is a heuristic and not an exhaustive search.

That limitation cannot account for the behaviour reported below. A more complete search would return a pattern of still lower Φ , never a higher one, and every departure from the truth reported here is a pattern whose Φ is already lower than the truth attains. Searching harder would widen that gap rather than close it. The over-fitting described in Section 10.6 is thus a property of the criterion and not an artefact of the procedure that minimises it.

10.3 What can be compared with least squares, and what cannot

For $Y = AX$ with Y and X of shape $N \times T$, minimising the sum of squared residuals gives the normal equations $YX' = AXX'$, hence

$$A = (YX') (XX')^{-1} \tag{10.2}$$

Only one of the two possible comparisons can be made here. Since $XX' = \sum_t x_t x_t'$ is a sum of T matrices of rank one, its rank is at most T whatever the values in X . When $T \geq N$ and X has full row

rank the inverse exists and (10.2) is the estimate; a comparison against it would then be meaningful. These illustrations use $N = 9$ and $T = 6$, so the rank is at most six out of nine and the matrix is singular for every X , not merely for unlucky ones. In the run reported below its three smallest singular values are 1.7×10^{-15} , 9.8×10^{-16} and 3.2×10^{-16} , which are zeros carrying rounding error. The least-squares estimate does not exist, and no comparison against it is possible.

What can be shown is the minimum-norm solution obtained from the pseudo-inverse. That is a different estimate: among the matrices reproducing the data exactly it selects the one of smallest Frobenius norm. Figure 1 show the linear least squares estimate of the matrix, using the block structured matrix and data as given in section A. For the block case it occupies all 81 cells, leaves a residual of 2×10^{-28} , and stands at Frobenius distance 4.02 from the true matrix.

-1.33	1.21	-0.64	0.87	-0.79	0.61	-0.31	-0.48	-0.44
-1.64	0.95	0.92	0.25	-0.25	0.36	-0.58	-0.29	-0.02
-1.78	0.67	0.86	0.26	-0.17	0.23	-0.30	-0.25	-0.05
0.38	-0.21	0.64	-1.00	1.32	1.49	0.35	-0.12	-0.47
-0.14	-0.10	-0.21	1.18	0.33	-0.39	0.08	0.33	0.54
0.16	-0.33	0.41	-1.37	-0.70	-0.53	-0.03	-0.21	-0.26
0.36	0.35	0.55	0.32	-0.04	-0.35	-0.96	0.70	0.34
-0.06	-0.84	-0.16	0.26	0.53	-0.27	0.42	1.00	-0.32
-0.12	0.28	-0.25	-0.41	-0.37	0.42	-0.14	.	-0.64

Figure 1: Linear Least Squares Estimated Matrix

A comparison against (10.2) itself requires a separate run with $T \geq N$, which is not attempted here.

10.4 One realisation

The trial shown is the first of the forty, chosen before any results were seen. The true block matrix and the matrix recovered from it are Figures 2 and 3

-1.67	2.30	-1.30	.	.	.	.	.	.
-1.61	1.72	0.93	.	.	.	.	.	.
-1.75	1.19	0.77	.	.	.	.	.	.
.	.	.	-1.56	1.82	1.32	.	.	.
.	.	.	1.42	0.55	-0.72	.	.	.
.	.	.	-1.77	-0.42	-0.68	.	.	.
.	.	.	.	.	.	-1.44	0.81	0.98
.	.	.	.	.	.	1.63	1.86	0.44
.	.	.	.	.	.	-0.59	-0.66	-1.49

Figure 2: True Block Matrix

All 27 occupied cells are found, one further cell is occupied that should not be, and the estimated error is 0.0798 against a true 0.10. The Frobenius distance from the truth is 0.373, against 4.02 for the minimum-norm solution.

The blocks are visible in the recovered matrix, and nothing in the calculation was told to look for them.

-1.65	2.36	-1.35	.	.	.	.	.	.
-1.59	1.77	0.88	.	.	.	.	.	.
-1.79	1.22	0.74	.	.	.	.	.	.
.	.	.	-1.57	1.83	1.31	.	.	.
.	.	.	1.41	0.56	-0.72	.	.	.
.	.	.	-1.76	-0.38	-0.71	.	.	.
.	-0.24	.	.	.	.	-1.65	0.72	0.92
.	.	.	.	.	.	1.58	1.88	0.36
.	.	.	.	.	.	-0.60	-0.69	-1.49

Figure 3: Recovered Block Matrix

For the scattered case the true and recovered matrices are in Figures 4 and 5

.	.	-1.67	.	.	.	2.30	.	-1.30
.	-1.61	1.72	0.93	.	.	.	.	.
.	.	.	.	.	-1.75	1.19	.	0.77
.	-1.56	.	1.82	.	.	.	1.32	.
1.42	.	0.55	.	.	.	.	-0.72	.
-1.77	.	.	.	.	.	-0.42	-0.68	.
.	-1.44	0.81	.	.	.	.	0.98	.
.	.	.	.	1.63	1.86	.	0.44	.
.	.	.	.	-0.59	-0.66	-1.49	.	.

Figure 4: True Scattered Matrix

Here the recovery is exact: all 27 cells found, none added, none missed, with an estimated error of 0.0826.

One number deserves attention. In the block case Φ at the recovered pattern is 53.674, while Φ at the true pattern is 55.919. The criterion prefers the pattern with one spurious cell to the truth, by more than two natural units. The search did not fail; the criterion selected what it was asked to select.

10.5 Forty realisations

One trial cannot distinguish a property of block structure from an accident of one draw. The whole construction was therefore repeated over forty seeds, with the coefficient values, the inputs, the errors and the scattered pattern all redrawn each time.

.	.	-1.75	.	.	.	2.26	.	-1.27
.	-1.56	1.67	0.93	.	.	.	.	.
.	.	.	.	.	-1.72	1.24	.	0.82
.	-1.60	.	1.81	.	.	.	1.34	.
1.42	.	0.53	.	.	.	.	-0.71	.
-1.79	.	.	.	.	.	-0.47	-0.68	.
.	-1.51	0.87	.	.	.	.	0.98	.
.	.	.	.	1.60	1.93	.	0.50	.
.	.	.	.	-0.64	-0.65	-1.47	.	.

Figure 5: Recovered Scattered Matrix

case	correct of 27	false	missed	exact	median $\hat{\sigma}$
block	26.20	1.57	0.80	14 of 40	0.091
scatter	26.40	1.40	0.60	17 of 40	0.089

The two cases perform alike. Compared within each seed, the block case was the worse of the two seventeen times, the better eleven times, and equal twelve times; a sign test on the twenty-eight untied pairs gives $p = 0.35$, so the difference is within what chance produces.

That result is what the construction requires. The prior treats all cells alike and the rows are fitted separately, so no information can pass from the arrangement of the cells to the estimate. The method is indifferent to structure, which is what declining to impose structure means. It also follows that the method does not exploit block structure; it recovers such a matrix neither better nor worse than any other.

10.6 What the illustrations show

Three conclusions may be drawn, and a fourth qualification attaches to them.

The method recovers most of the correct cells. Across the forty trials it finds about 26.3 of the 27 occupied cells and misses fewer than one, in a setting where the unrestricted estimate does not exist at all.

It recovers the whole matrix exactly in about two cases in five, and otherwise occupies between one and two cells that should be empty. The estimated error is consistently a little below the truth, near 0.09 against 0.10, which is the mark of those extra cells absorbing part of the measurement error.

The residue is not distributed evenly. Repeating the forty trials with the floor lowered to $a_{\min} = 0.05$ raises the number of falsely occupied cells from 1.49 to 5.90 per matrix, and the additional cells sit at small magnitudes. Their median magnitude is 0.138, against 1.171 for the correctly occupied cells, and a third of them fall below 0.10, against fewer than one in two hundred of the correctly occupied cells.

Raising the floor to a value an analyst would assert removes most of them and brings the estimated error back to 0.10.

The qualification is that this concentration points at the approximation of Section 9. Equation (9.2) treats $\prod 1/|A_{ij}|$ as constant across the region of good fit and takes no account of the truncation at $a_{\min}$. Both are least accurate for coefficients close to the floor, and both err in the direction of overstating the integral there, which is exactly where the spurious cells appear. Handling the truncation properly, rather than by the Gaussian approximation, is the first improvement to attempt.

11 What is not settled here

Three matters are deliberately left open.

The search over patterns is not addressed. There are 2^q of them, and q may be forty thousand. Equation (7.4) says how to score a pattern; it does not say how to find a good one. Note that, before p and σ are removed, the N rows of A are separate problems, each involving only its own row of Y . The rows are connected solely through the two shared quantities. That separation is the natural starting point for a search procedure, and it is the whole of the hierarchical structure in the present model.

The second point is that one p and one σ serve the entire matrix. In the AutoClass formalism, parameters are shared within classes rather than across everything. Allowing p and σ to vary by row, or by group of rows, is the natural next step and will be taken up separately.

Third, the inputs are taken to be exact. Error in X as well as Y is a different and harder problem and is not considered.

12 Verification

Every algebraic step above was checked with Maxima. The script is `maxima/acv_prelim.mac` in this directory, and it verifies:

1. the divergence of $\int da/|a|$ at both ends, equations (4.3) and (4.4);
2. that the bounded densities (4.6) and (5.8) integrate to unity;
3. that L_A is unchanged by a change of units;
4. the integral over p , equation (6.2), for general α and β ; its reduction (6.3) at $\alpha = \beta = 1$, in all three of the forms written there, symbolically and at particular values of q and M ; the ratio (6.5) and hence the increment (6.6); and the posterior mean of p ;

5. the closed form (6.8) for the integral over σ , checked against numerical quadrature at three sets of values;
6. the finite limit (6.10) as $R \rightarrow 0$, and the divergence of the corresponding unbounded expression;
7. the location (5.5) of the maximum of the integrand in (5.4), the unbounded growth of its height as $R \rightarrow 0$, and the closed form (5.6), the last checked against numerical quadrature over $(0, \infty)$ at $R = 1, 0.1, 0.01, 0.001$;
8. the stationarity condition (8.2);
9. the Gaussian integral behind (9.2), and the closed form (9.4) for the integral over σ with the exponent $M - n - 1$, checked against numerical quadrature at three values of ν and confirmed to reduce to (6.8) at $M = 0$.

The illustrations of Section 10 are produced by the scripts in the `examples` directory. `recover.py` carries the first trial together with the comparison between the criterion of Section 7 and that of Section 9; `sweep.py` repeats the construction over many seeds; `final.py` produces the numbers and the tables printed above. The search is implemented twice, independently, and the two implementations agree on the scattered case of the first seed. That agreement is the check each was written against, and it is what exposed an error in the first: it collected candidate patterns under a key that recorded only how many cells each row received, so that only the first value of σ producing a given allocation contributed a pattern, and part of the candidate set went unexamined.

A Numerical Examples

Every number needed to repeat the calculations of Section 10 is printed here: the two matrices to be recovered, the inputs, and the outputs of both cases. A reader who works from these tables alone reaches the same results, and needs neither our programs nor the generator that produced the figures.

The arrays were nonetheless produced from a single stream of pseudo-random numbers, started at seed 20260726 and drawn in the order given below, using the default generator of the `numpy` library, version 2.5. The script `examples/final.py` carries out all of it. That recipe is recorded for anyone wishing to extend the example to other draws; it is not needed to check the one reported.

1. Twenty-seven magnitudes are drawn, each with log uniform on $[\log 0.40, \log 3.00]$.
2. Twenty-seven signs are drawn, positive and negative equally likely.
3. Each magnitude is multiplied by its sign and rounded to two decimal places. These twenty-seven values are the occupied cells of both matrices.
4. The values are placed row by row into three blocks of size three along the diagonal, giving the first matrix.
5. For the second matrix, each row keeps its own three values and moves them to three columns drawn without replacement from the nine. A row that happens to draw its own block columns causes the whole arrangement to be drawn again.
6. The nine by six array of inputs X is drawn from the standard normal.
7. A nine by six array of measurement errors is drawn from the standard normal and multiplied by $\sigma = 0.10$.

The outputs are then $Y = AX + E$, formed separately for each of the two matrices from the same X and the same E .

A.1 The block matrix

The occupied cells lie in three blocks of size three along the diagonal. Empty cells are shown as a dot.

-1.67	2.30	-1.30	.	.	.	.	.	.
-1.61	1.72	0.93	.	.	.	.	.	.
-1.75	1.19	0.77	.	.	.	.	.	.
.	.	.	-1.56	1.82	1.32	.	.	.
.	.	.	1.42	0.55	-0.72	.	.	.
.	.	.	-1.77	-0.42	-0.68	.	.	.
.	.	.	.	.	.	-1.44	0.81	0.98
.	.	.	.	.	.	1.63	1.86	0.44
.	.	.	.	.	.	-0.59	-0.66	-1.49

A.2 The scattered matrix

The same twenty-seven values, each row keeping its own three, moved to columns drawn at random.

.	.	-1.67	.	.	.	2.30	.	-1.30
.	-1.61	1.72	0.93	.	.	.	.	.
.	.	.	.	.	-1.75	1.19	.	0.77
.	-1.56	.	1.82	.	.	.	1.32	.
1.42	.	0.55	.	.	.	.	-0.72	.
-1.77	.	.	.	.	.	-0.42	-0.68	.
.	-1.44	0.81	.	.	.	.	0.98	.
.	.	.	.	1.63	1.86	.	0.44	.
.	.	.	.	-0.59	-0.66	-1.49	.	.

A.3 The inputs

The nine by six array X . Both cases use it unchanged, so it is given once. The row index j runs over the components of the input vector and the column index t over the six observations.

	$t = 1$	$t = 2$	$t = 3$	$t = 4$	$t = 5$	$t = 6$
$j = 1$	-0.166450	-0.721720	-1.956128	0.498841	-1.404545	-0.285642
$j = 2$	0.516260	0.647030	-0.876677	-0.527945	0.386305	1.130293
$j = 3$	-1.164318	1.145200	0.142888	-1.157197	-0.554896	-1.233712
$j = 4$	1.711768	-0.854384	-0.920882	0.134728	2.363427	-0.971539
$j = 5$	-1.785136	-0.735074	-0.423137	-0.314053	1.523255	-1.244201
$j = 6$	-1.925274	-0.040422	-0.242413	-0.164816	1.433408	1.335336
$j = 7$	-0.580635	-1.296970	0.915671	-0.097830	0.186884	0.458883
$j = 8$	0.030004	0.803816	0.392157	1.644819	1.237441	-2.244700
$j = 9$	0.844496	-0.704454	-1.322737	-1.225774	-0.544313	-0.788596

A.4 The outputs

The nine by six array Y for the block case, formed as $Y = AX + E$ with A the matrix of Section A.1 and E the measurement errors. The row index i runs over the components of the output vector.

	$t = 1$	$t = 2$	$t = 3$	$t = 4$	$t = 5$	$t = 6$
$i = 1$	3.089032	1.152337	0.945229	-0.521665	4.004424	4.778189
$i = 2$	0.160786	3.342429	1.686812	-2.720215	2.428925	1.354611
$i = 3$	0.066173	2.819019	2.494934	-2.510754	2.645237	0.923847
$i = 4$	-8.470004	-0.074040	0.441555	-0.879926	0.977153	0.986482
$i = 5$	2.800136	-1.651273	-1.298047	0.206664	3.134637	-3.015192
$i = 6$	-0.948718	1.893286	1.935759	-0.126420	-5.763637	1.223792
$i = 7$	1.646148	1.947265	-2.188468	0.302910	-0.001640	-3.394931
$i = 8$	-0.701104	-0.797361	1.623413	2.503228	2.384561	-3.805208
$i = 9$	-0.854054	1.217295	1.098340	0.871725	-0.169908	2.499949

The same for the scattered case, using the matrix of Section A.2 and the same E :

	$t = 1$	$t = 2$	$t = 3$	$t = 4$	$t = 5$	$t = 6$
$i = 1$	-0.378846	-4.032070	3.467584	3.322333	2.113083	4.238572
$i = 2$	-1.154212	0.135980	0.713241	-0.947912	0.640831	-4.747311
$i = 3$	3.385596	-2.110847	0.500295	-0.890321	-2.550343	-2.369103
$i = 4$	2.340323	-1.518996	0.304329	3.359343	5.331900	-6.521777
$i = 5$	-0.933285	-1.036591	-2.913672	-1.042962	-3.217771	0.542177
$i = 6$	0.540249	1.320387	2.774337	-2.092267	1.600207	1.728883
$i = 7$	-1.698984	0.902531	1.871200	1.465823	0.005519	-4.969843
$i = 8$	-6.659635	-0.788115	-0.984590	0.047925	5.711317	-0.563054
$i = 9$	3.270531	2.325823	-1.028175	0.513020	-2.177188	-0.716799

The errors themselves are not printed. They are the difference between these outputs and AX , and nothing in the calculation uses them: the method sees only X and Y .

A.5 The precision of the printed values

The coefficients of A are exact as printed, since the generator rounds them to two decimal places. The entries of X and Y are given to six decimal places, so a reader recomputing from them works with slightly different numbers from those used above.

That rounding does not affect the result. Repeating the whole calculation on the rounded arrays recovers the same set of occupied cells in both cases, and the estimated coefficients differ from those reported by less than 10^{-6} . Six decimal places are therefore sufficient, and the check is carried out by `examples/final.py` rather than asserted.